

HYPOTHESIS TESTING

Likelihood ratio tests

Likelihood ratio tests

- The Neyman-Pearson Lemma gives a method for most powerful testing to test between two simple hypotheses.
- Extending that general idea, likelihood ratio tests are a great “general purpose” method for testing, applicable in many situations which might otherwise be intractable (or at least, very difficult).

Parameter space

- The parameter space (denoted by Ω) is the set of all possible values of a parameter (or set of parameter).

- Example:

- binomial(p) distribution
- Poisson(λ) distribution
- $N(\mu, \sigma^2)$ distribution
- Gamma(α, β) distribution

$p = \text{parameter}$ $\Omega = \{p : 0 < p < 1\}$
 $\Omega = (0, 1)$

$\Omega = \{\lambda : \lambda > 0\} = (0, \infty)$

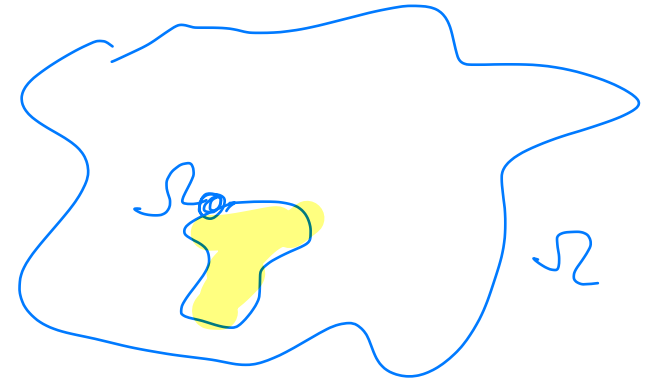
(c) $\Omega = \{(\mu, \sigma^2) : -\infty < \mu < \infty, \sigma^2 > 0\}$
 $= (-\infty, \infty) \times (0, \infty)$



(d) $\alpha > 0, \beta > 0$

$\Omega = \{(\alpha, \beta) : \alpha > 0, \beta > 0\}$

Restricted parameter space



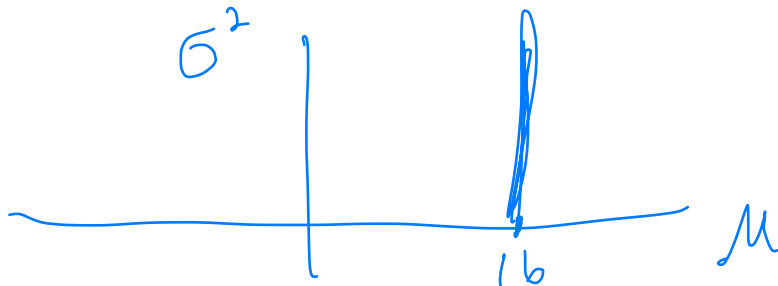
$$H_0: \theta \in \Omega_0$$

- The restricted parameter space (denoted by Ω_0) is the set of all possible values of a parameter (or set of parameters) *restricted* by the null hypothesis H_0 .

- Example again:

- binomial(p) distribution $H_0: p = \frac{1}{2}$; $H_a: p \neq \frac{1}{2}$ $\Omega_0 = \{\frac{1}{2}\}$ $\Omega = (0, 1)$
- Poisson(λ) distribution $H_0: \lambda \leq 2$; $H_a: \lambda > 2$ $\Omega_0 = [0, 2]$
- $N(\mu, \sigma^2)$ distribution $H_0: \mu = 16$; $H_a: \mu \neq 16$
- Gamma(α, β) distribution $H_0: \alpha \leq 1$; $H_a: \alpha > 1$

$$(c) \Omega_0 = \{(\mu, \sigma^2): \mu = 16, \sigma^2 > 0\}$$



$$(d) \Omega_0 = \{(\alpha, \beta): \alpha \leq 1, \beta > 0\}$$

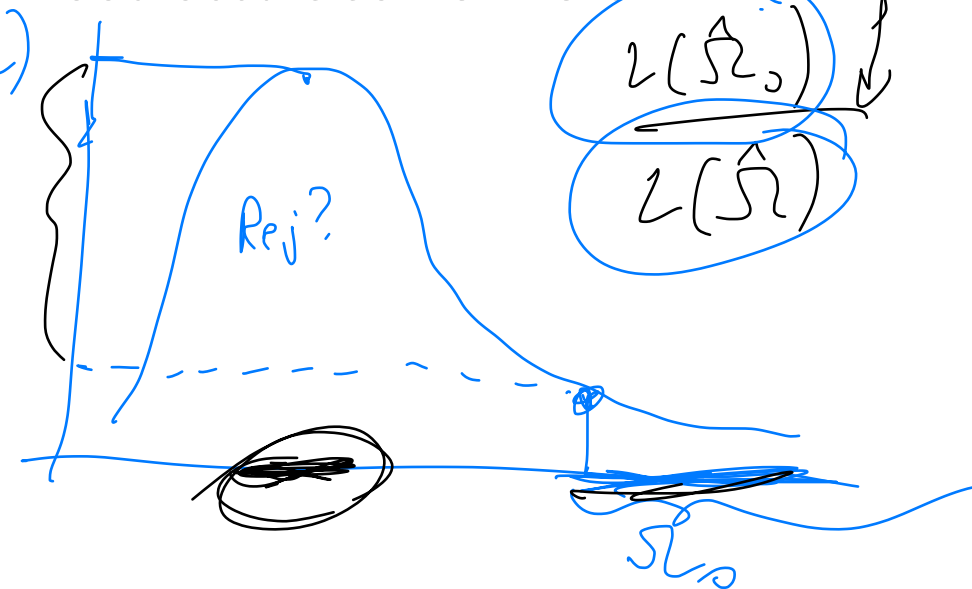
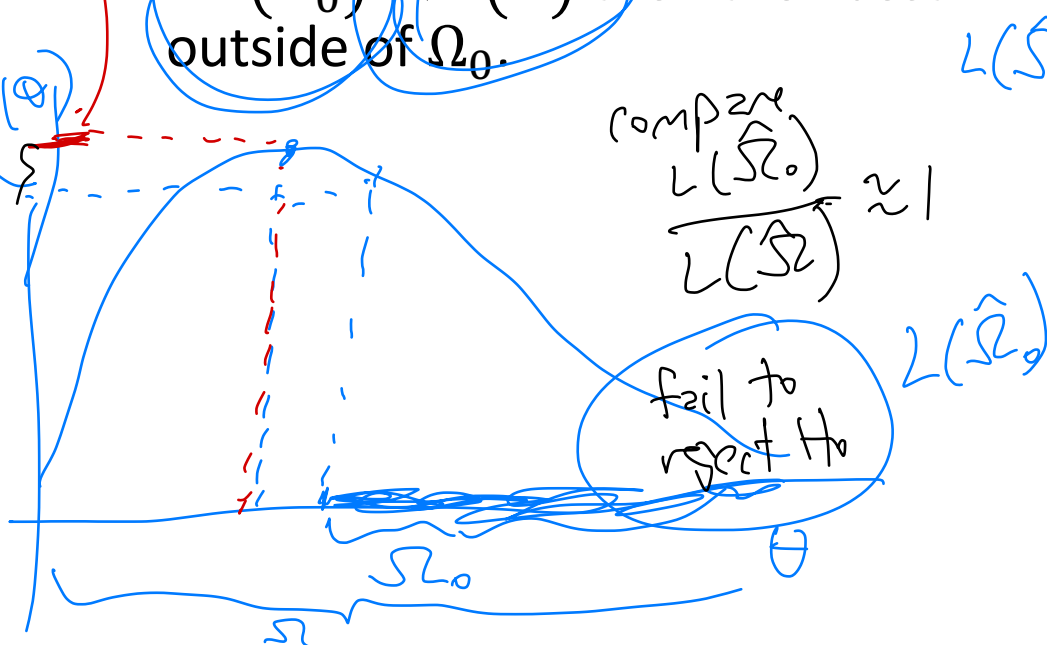
$L(\hat{\Omega}_0)$ $L(\hat{\Omega})$

Restricted and unrestricted likelihood functions

- Denote by $L(\hat{\Omega})$ the likelihood function, maximized over Ω .
- Denote by $L(\hat{\Omega}_0)$ the likelihood function, maximized over Ω_0 .
- The likelihood ratio test compares these.
- If $L(\hat{\Omega}_0) = L(\hat{\Omega})$ then the "best" likelihood is compatible with H_0 .
- If $L(\hat{\Omega}_0) < L(\hat{\Omega})$ then the "best" likelihood occurs somewhere outside of Ω_0 .

Small

~~Small~~



Likelihood ratio test

- A likelihood ratio test is based on comparing $L(\hat{\Omega}_0)$ to $L(\hat{\Omega})$:

$$\lambda = \frac{L(\hat{\Omega}_0)}{L(\hat{\Omega})} = \frac{\max_{\theta \in \Omega_0} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$$

- The rejection region is determined by $\lambda \leq k$, where k is chosen to ensure the chosen Type I error rate.

Reject H_0 if $\lambda \leq k$

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}y^2}$$

$$L(\sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}y_i^2}$$

$$\Omega_0 = \{\sigma^2 : \sigma^2 = 1\} = \{1\}$$

$$\Omega = \{\sigma^2 : \sigma^2 > 0\}$$

Likelihood ratio test $\mu = 0$

- Example 7: $Y_1, Y_2, \dots, Y_n \sim iid N(0, \sigma^2)$ with $H_0: \sigma^2 = 1$ and $H_a: \sigma^2 > 1$. Find the likelihood ratio test.

$$L(\sigma^2) = (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2} \sum Y_i^2}$$

Restricted - $\max L(\sigma^2)$ over Ω_0

$$L(1) = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2} \sum Y_i^2}$$

✓ maximize over Ω ...

$$\hat{\sigma}^2 = \frac{1}{n} \sum Y_i^2$$

unrestricted MLE

$$L(\hat{\sigma}^2) = (2\pi)^{-\frac{n}{2}} (\hat{\sigma}^2)^{-\frac{n}{2}}$$

skipped steps

$$e^{-\frac{1}{2\hat{\sigma}^2} \sum Y_i^2}$$

$$\sum Y_i^2 = n \hat{\sigma}^2$$

$$= (2\pi)^{-\frac{n}{2}} (\hat{\sigma}^2)^{-\frac{n}{2}} e^{-\frac{1}{2\hat{\sigma}^2} n \hat{\sigma}^2} = (2\pi)^{-\frac{n}{2}} (\hat{\sigma}^2)^{-\frac{n}{2}} e^{-\frac{n}{2}}$$

Likelihood ratio $\frac{L(1)}{L(\hat{\sigma}^2)} = \frac{(2\pi)^{-\frac{n}{2}} e^{-\frac{n}{2}\hat{\sigma}^2}}{(2\pi)^{-\frac{n}{2}} (\hat{\sigma}^2)^{-\frac{n}{2}} e^{-\frac{n}{2}}} = (\hat{\sigma}^2)^{\frac{n}{2}} e^{-\frac{n}{2}(\hat{\sigma}^2 - 1)}$

Reject H_0 if $(\hat{\sigma}^2)^{\frac{n}{2}} e^{-\frac{n}{2}(\hat{\sigma}^2 - 1)} \leq k$

if $\frac{n}{2} \log(\hat{\sigma}^2) - \frac{n}{2}(\hat{\sigma}^2 - 1) \leq \log k$

if $\log(\hat{\sigma}^2) - (\hat{\sigma}^2 - 1) \leq \frac{2 \log k}{n}$

$\log z - (z-1)$

$\frac{d}{dz} \log z - (z-1) = \frac{1}{z} - 1 < 0$ for $z > 1$

$\hat{\sigma}^2 > ?$

$\frac{d}{dz} \log z - (z-1) \downarrow$ (if $z > 1$)

As $z \uparrow$

If $\hat{\sigma}^2$ is "large", $\log(\hat{\sigma}^2) - (\hat{\sigma}^2 - 1)$ is "small"

Reject H_0 if $\hat{\sigma}^2$ large $\Leftrightarrow$ Reject H_0 if $\uparrow$

Reject H_0 if $\hat{\sigma}^2$ is large

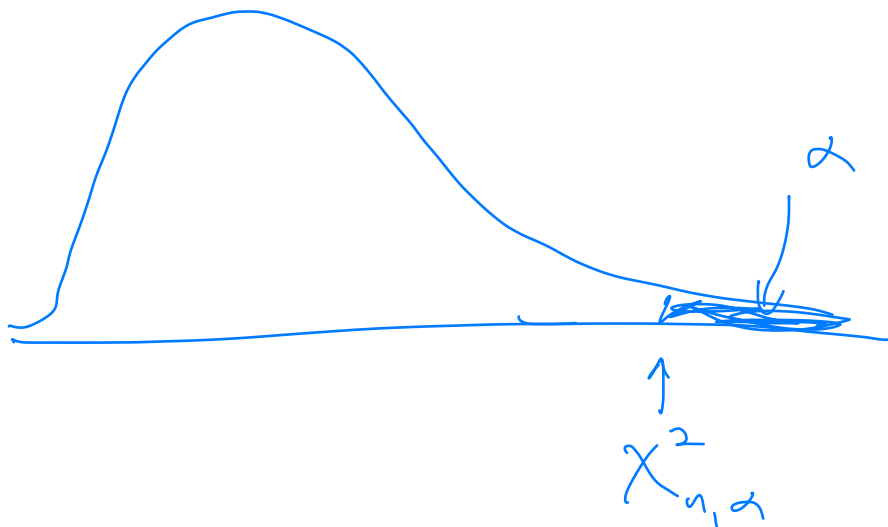
want $P(\text{Reject } H_0) = \alpha$ if H_0 is true

$$P(\hat{\sigma}^2 > c) = \alpha \quad \text{if } H_0 \text{ is true}$$

$$P\left(\frac{\sum Y_i^2}{n} > c\right) = \alpha$$

$$P(\sum Y_i^2 > cn) = \alpha$$

$$\begin{aligned} \hookrightarrow \sigma^2 = 1 &\Rightarrow Y_1, Y_2, \dots, Y_n \\ Y_i^2 &\sim \chi_1^2 \Rightarrow Y_i \sim N(0, 1) \\ \sum Y_i^2 &\sim \chi_n^2 \end{aligned}$$



Reject H_0 if
 $\sum Y_i^2 > \chi_{n, \alpha}^2$